Neonatal Seizure

126

Introduction

- ☑ Seizures in the newborn represent the most distinctive common manifestations of neurological disease in the neonatal period. The incidence increase with lower gestational age and birth weight and is most common in the very low birth weight (VLBW) infant.
- ☑ Estimated incidences are up to 58/100 live births in the VLBW infant and 1 to 3.5/1000 live births in the term infant.
- ☑ Most seizures during the newborn period are acute symptomatic seizures due to cerebral injury or dysfunction due to varied etiology.
- ☑ Important manifestation to alert the clinician to underlying neurological disorders.
- ☑ Seizure onset and semiology will guide in identifying possible etiology and determining appropriate management.
- ☑ Amplitude-integrated electroencephalography (aEEG) can pick up 70–80% of seizures in monitoring.

Clinical Versus Electrical Seizures

- ☑ As per the World Health Organization (WHO) recommendations, all clinically apparent seizures lasting for >3 minutes or brief serial seizures are to be treated.
- ☑ If continuous electroencephalogram (EEG) monitoring is available, all electrical seizures should be treated even in absence of clinically apparent seizures, especially if babies are paralyzed.

Types of Neonatal Seizures

- ☑ Electrical activity that evolves over time and meets criteria
- ☑ *Electroclinical seizure:* Electrographic seizure with associated clinical signs
- ☑ *EEG-only (subclinical, nonconvulsive, and occult) seizures:* Electrographic seizures without clinical signs

Electrographic Criteria for Neonatal Seizures

- ☑ Sudden change in EEG
- ☑ Repetitive waveforms that evolve in morphology, frequency, and/or location
- ☑ *Amplitude:* At least 2 µV
- ☑ *Duration*: At least 10 seconds
- ☑ Seizures must be separated by at least 10 seconds to be considered separate
- ☑ Clinical signs may or may not be present
- ☑ EEG seizures can be as follows:
  - Unifocal—seizures arise from a single region
  - Multifocal—seizures originate from at least three independent foci with at least one in each hemisphere
  - Lateralized—seizures propagate within a single hemisphere
  - Bilateral independent—seizures occur simultaneously in two regions and begin, evolve, and behave independently
  - Bilateral—both hemispheres involved
  - Migrating—the seizure moves sequentially from one hemisphere to another
  - Diffuse—asynchronous involvement of all brain regions

Nonepileptic Phenomenon

- ☑ Roving eye movement, nystagmoid jerks, sucking, and other limb movement during sleep or in drowsy state
- ☑ May be mistaken as seizure activity
- ☑ Movement stops with gentle restrain, and no autonomic phenomena occurs
- ☑ Causes could be mild hypoxic-ischemic encephalopathy (HIE)/metabolic/drug withdrawal

Etiology

Etiologies of neonatal seizure are given in **Box 1**.

BOX 1: Common underlying etiologies (responsible for 80–85% of seizure).

- ☑ Hypoxic-ischemic encephalopathy (38%)
- ☑ Stroke (18%)
- ☑ Intracranial hemorrhage (11%)
- ☑ Intracranial infections (5%)
- ☑ Cerebral dysgenesis (4%)
- ☑ Metabolic and genetic (12–15%)

Classification of Neonatal Seizure

- ☑ Subtle seizure
- ☑ Clonic seizure
- ☑ Tonic seizure
- ☑ Spasm
- ☑ Myoclonic seizure
- ☑ Term versus preterm **(Table 1)**

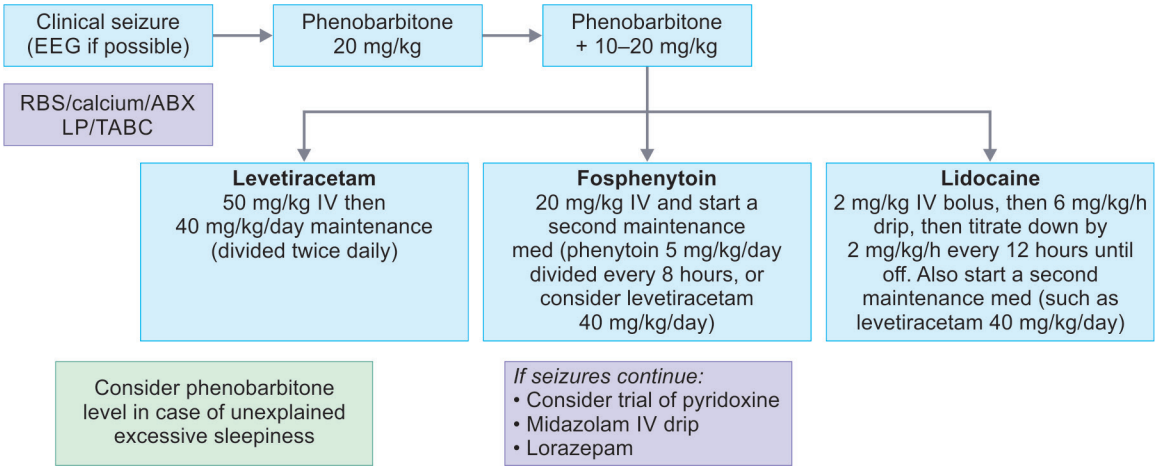
TABLE 1: Term versus preterm neonatal seizure.

Cause	*Preterm*	*Full term*
Hypoxic-ischemic encephalopathy (HIE)	+++	+++
Intracranial hemorrhage	++	+
Intracranial infection	++	++
Development defect	++	++
Hypoglycemia	+	+
Hypocalcemia	+	+
Epilepsy syndrome	–	+

Management

Management of neonatal seizure is depicted in **Flowchart 1**.

Flowchart 1: Management of neonatal seizure.



(ABX: antibiotics; EEG: electroencephalogram; LP: lumbar puncture; RBS: random blood sugar; TABC: temperature/airway/breathing/circulation)

How to Plan to Wean the Anticonvulsants if no Clinical Seizure or EEG Awaited

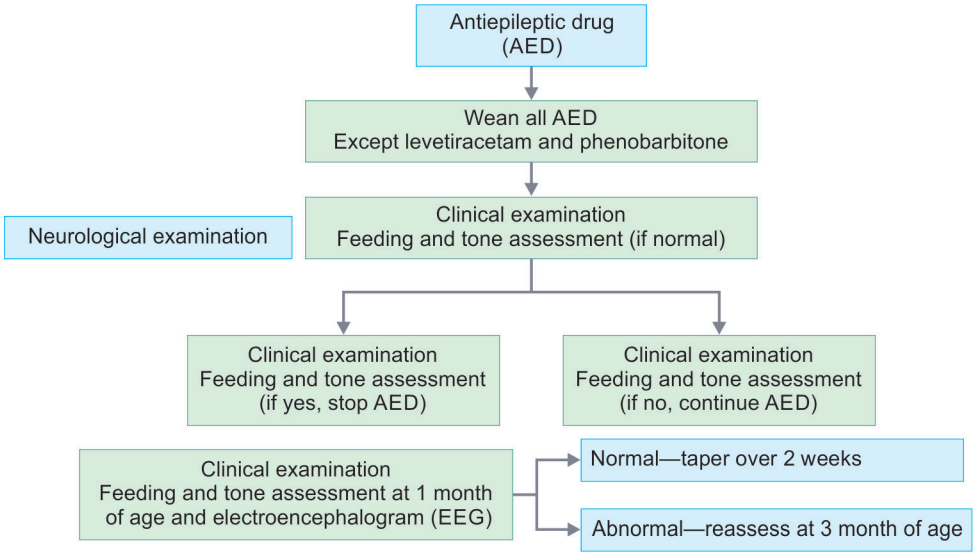
- ☑ Monitor EEG (if available)
- ☑ If on phenobarbitone (PB) maintenance (get trough level, if available)
- ☑ Get MRI brain/lumbar puncture and other relevant investigations to rule out the cause of seizure
- ☑ Always attempt to wean to one drug for maintenance
- ☑ Try to stop all medication before discharge if seizure free for 48–72 hours

Determinants of Duration and When to Stop Anticonvulsant Therapy (Flowchart 2)

It is mainly depends on three factors:

1. Neurological examination at the time of discharge—examine for feeding pattern and tone assessment
2. Cause of neonatal seizure
3. EEG pattern

Flowchart 2: Weaning of all AEDs.



Further Reading

☑ Volpe J. Neonatal seizure. In: Volpe J, Inder T, Darras B, de Vries L, du Plessis A, Neil J, Perlman J (Eds). Volpe's Neurology of the Newborn, 6th edition. Amsterdam, Netherlands: Elsevier; 2017.

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